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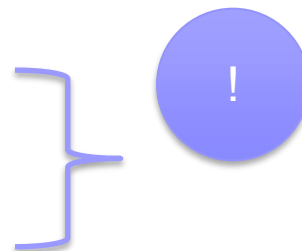
RDF & SPARQL: Exercises Exercises



dat•AI data * AI
AI * data

Overview

- Recap on RDF
- Recap on SPARQL



Indicates content that is more interesting from a pragmatic point of view and/or novel w.r.t. theory slides

Separate presentation

- Exercises in RDF + SPARQL
 - For RDF exercise this year also consider
 - DS2526 - ESE1.0 - Exercises about RDF modeling
- Exercises in Advanced SPARQL

Exercises

Exercise no. 1 // RDF

- Exercise #1 from “Knowledge Representation and Reasoning – Mod. 2 - Exercise Sheet 3 & 4: RDF & SPARQL”
 - In class: warm-up

Exercise no. 2 // RDF

- Describe in RDF the following statements:
 - Dipartimento di Informatica, Sistemistica e Comunicazione of Università degli studi di Milano – Bicocca is located in Viale Sarca 336, Milan
 - The Web site <http://disco.unimib.it> contains the activity of Dipartimento di Informatica, Sistemistica e Comunicazione
 - (a) Write these statements in RDF in N3, **Turtle** and RDF/XML format without blank nodes, using at least three triples for each statement.
 - (b) Write these statements in RDF in N3, **Turtle** and RDF/XML format with blank nodes, using at least three triples for each statement.
 - (c) Draw the graph of the exercise.

Possible solution in N3 (a)

NO

<http://disco.unimib.it/disco>

<http://disco.unimib.it/name>

“Dipartimento di Informatica, Sistemistica e Comunicazione”@it .

<http://disco.unimib.it/disco>

<http://disco.unimib.it/address>

“Viale Sarca 336, Milan”@it .

<http://disco.unimib.it/disco>

<http://disco.unimib.it/partOf> <http://disco.unimib.it/unimib> .

Possible solution N3 (b)

NO

Subject	Predicate	Object
<http://disco.unimib.it/disco>	<http://disco.unimib.it/address>	_:A231887eeX6aX143a880X4aXX6dX9fff
<http://disco.unimib.it/disco>	<http://disco.unimib.it/name>	"Dipartimento di Informatica, Sistemistica e Comunicazione"@it
<http://disco.unimib.it/disco>	<http://disco.unimib.it/department>	<http://disco.unimib.it/unimib>
_:A231887eeX6aX143a880X4aXX6dX9fff	<http://disco.unimib.it/cap>	"20126" ^^<http://www.w3.org/2001/XMLSchema#integer>
_:A231887eeX6aX143a880X4aXX6dX9fff	<http://disco.unimib.it/country>	<http://disco.unimib.it/Italia>
_:A231887eeX6aX143a880X4aXX6dX9fff	<http://disco.unimib.it/street>	"Viale Sarca, 336"@it
_:A231887eeX6aX143a880X4aXX6dX9fff	<http://disco.unimib.it/city>	<http://disco.unimib.it/Milano>
<http://disco.unimib.it/unimib>	<http://disco.unimib.it/name>	"Università degli studi di Milano-Bicocca"@it

Possible solution Turtle (a)

@prefix ex: <<http://disco.unimib.it#>> .

ex:disco ex:name "Dipartimento di Informatica Sistemistica e Comunicazione" .

ex:disco ex:department ex:Bicocca .

ex:Bicocca ex:name "Università degli Studi di Milano – Bicocca" .

ex:disco ex:address ex:addressdisco .

ex:addressdisco ex:street "Viale Sarca, 336" .

ex:addressdisco ex:CAP "20126" .

ex:addressdisco ex:municipality "Milano" .

ex:addressdisco ex:country "Italia" .

@prefix rdf: <<http://www.w3.org/1999/02/22-rdf-syntax-ns#>> .

@prefix : <<http://example.com#>> .

<http://disco.unimib.it> rdf:type :website .

<http://disco.unimib.it> :hosts :disco-activity .

:disco-activity :has-homepage :disco-activity-homepage .

:disco-activity-homepage rdf:type :homepage .

:disco-activity-homepage :url <<http://disco.unimib.it/>> .

Possible solution Turtle (b)

@prefix ex: <http://disco.unimib.it#> .

ex:disco ex:name "Dipartimento di Informatica, Sistemistica e Comunicazione" .

ex:disco ex:department ex:unimib .

ex:unimib ex:name "Università degli studi di Milano – Bicocca" .

ex:disco ex:address [ex:street "Sarca, 336";

ex:CAP "20126";

ex:municipality "Milano";

ex:country "Italia"] .

@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .

@prefix : <http://www.example.com#> .

<http://disco.unimib.it.org/> rdf:type :website .

<http://www.disco.unimib.it> :hosts :disco-activity .

:disco-activity :has-homepage [rdf:type :homepage;

url "http://disco.unimib.it"] .

Possible solution RDF/XML (c)

NO

```
<?xml version="1.0"?>
```

```
<rdf:RDF
```

```
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
```

```
  xmlns:ex="http://disco.unimib.it/">
```

```
<rdf:Description rdf:about="http://disco.unimib.it/disco">
```

```
  <ex:name>"Dipartimento di Informatica, Sistemistica e Comunicazione"</ex:name>
```

```
  <ex:department> rdf:resource="http://disco.unimib.it/unimib">
```

```
  <ex:address rdf:nodeID="_g0"/>
```

```
</rdf:Description>
```

```
<rdf:Description rdf:nodeID="_g0">
```

```
  <ex:street>"Viale Sarca, 336"</ex:street>
```

```
  <ex:CAP>"20126"</ex:CAP>
```

```
  <ex:municipality>"Milano"</ex:municipality>
```

```
  <ex:country>"Italia"</ex:country>
```

```
</rdf:Description>
```

```
<rdf:Description rdf:about="http://disco.unimib.it/unimib">
```

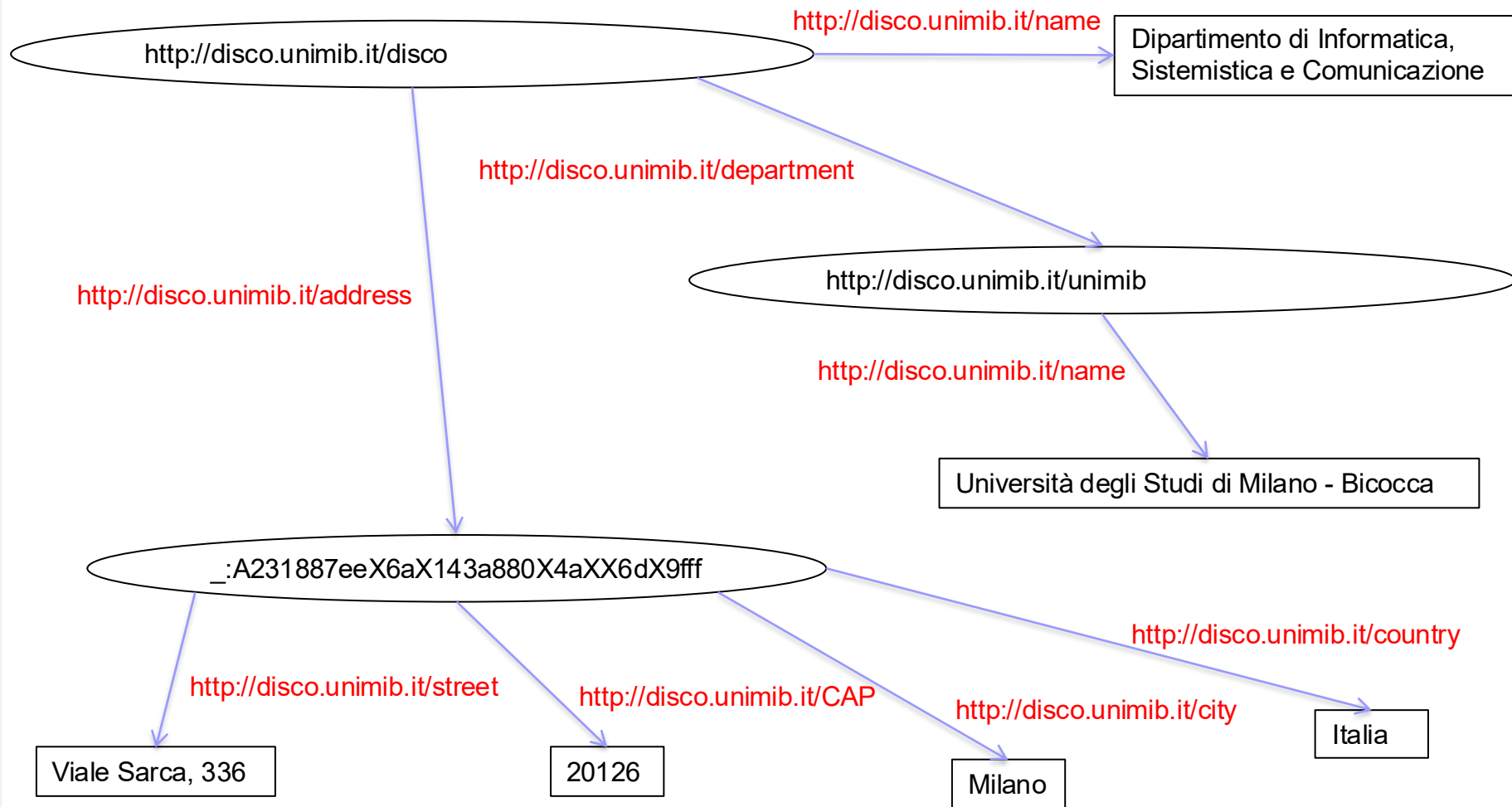
```
  <ex:name>"Università degli studi di Milano – Bicocca"</ex:name>
```

```
</rdf:Description>
```

```
</rdf:RDF>
```

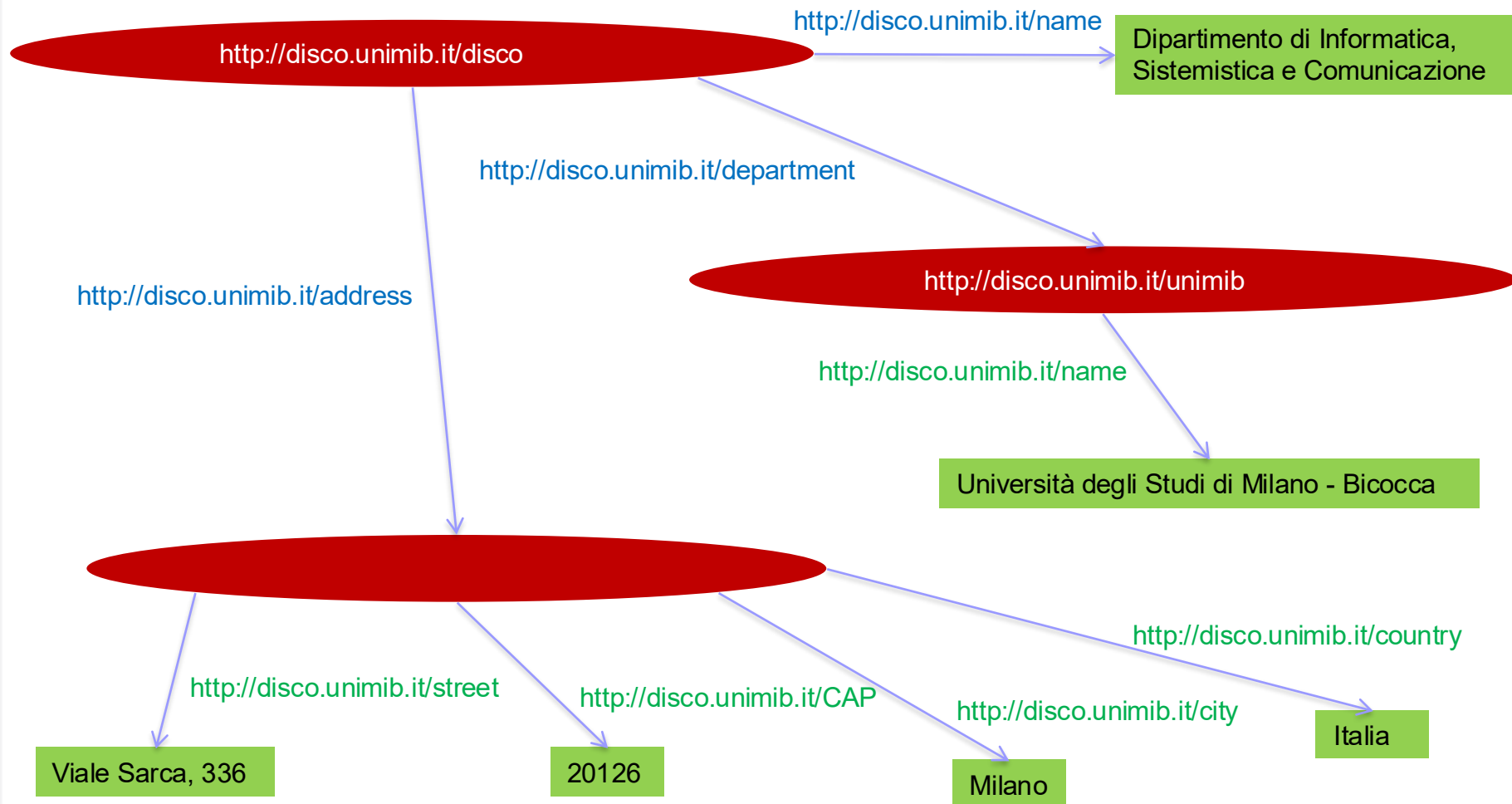
RDF Graph for (c)

Drawing the graph is very useful to learn to represent using a graph model



RDF Graph for (c)

Colors highlight literals (in green) and distinguish edges connecting IRI/BNs to IRI/BNs vs IRI/BNd to literals



Exercise no. 3 // RDF and Modeling

- Exercise #3 from “Knowledge Representation and Reasoning – Mod. 2 -Exercise Sheet 3 & 4: RDF & SPARQL”
 - Some exercise with reification
 - useful for practicing with blank nodes and the triple structure
 - ~ treat a statement as an entity
 - E.g., I believe that Marco is a nice guy
 - I believe A, A = Marco is a nice guy
 - Do the best you can / think how to do it in the given time
 - Rest of the exercise at home
- **Solutions in the file**

Exercise no. 4 // RDF and Modeling

- Exercise #2 from “Knowledge Representation and Reasoning – Mod. 2 - Exercise Sheet 3 & 4: RDF & SPARQL”
 - ☐ As before: some exercise with reification
 - ☐ Do the best you can / think how to do it in the given time
 - ☐ Discussion
 - ☐ Rest of the exercise at home
- **Solutions in the file**

Exercise no. 5 // RDF and Modeling

- Exercise #4 from “Knowledge Representation and Reasoning – Mod. 2 - Exercise Sheet 3 & 4: RDF & SPARQL”
- **Solutions in the file**

Exercises no. 6 // SPARQL Details

- Given the query in natural language write the corresponding SPARQL query
- Use as endpoint
<https://dbpedia.org/sparql>

How many employees does Google have?

Select Queries

- How many employees does Google have?

```
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX dbo: <http://dbpedia.org/ontology/>
SELECT ?num
WHERE {
    dbr:Google dbo:numberOfEmployees ?num .
}
```

Information
removed from
DBpedia

Results:

num
"57100"^^<http://www.w3.org/2001/XMLSchema#nonNegativeInteger>

How Did I Debugged the Query

- Is the property a valid property?
 - I searched for dbo:numberOfEmployees and it exists:
 - <https://dbpedia.org/ontology/numberOfEmployees>
- If the property is valid, two options:
 - It is deprecated and not used anymore
 - It is used but not with Google
 - How to find it out:
 - With a query (!)
 - PREFIX dbr: <http://dbpedia.org/resource/>
 - PREFIX dbo: <http://dbpedia.org/ontology/>
 - SELECT ?company ?num
 - WHERE {?company dbo:numberOfEmployees ?num . }
 - There are results... then I created a query with some results

How many employees does Google Rolex have?

```
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX dbo: <http://dbpedia.org/ontology/>
SELECT ?num
WHERE {
    dbr:Rolex dbo:numberOfEmployees ?num .
}
```

num
"7000"^^<http://www.w3.org/2001/XMLSchema#nonNegativeInteger>

In which country does the Nile
start?

Select Queries

- In which country does the Nile start?

PREFIX dbr: <<http://dbpedia.org/resource/>>

PREFIX dbp: <<http://dbpedia.org/property/>>

PREFIX dbo: <<http://dbpedia.org/ontology/>>

SELECT DISTINCT ?uri

WHERE {

dbr:Nile dbp:source2Location ?uri .

}

uri
http://dbpedia.org/resource/Rwanda
http://dbpedia.org/resource/Ethiopia

Results

Give me a list of all trumpet players
that were bandleaders.

Select Queries

- Give me a list of all trumpet players that were bandleaders.

```
PREFIX dbo: <http://dbpedia.org/ontology/>
```

```
PREFIX dbr: <http://dbpedia.org/resource/>
```

```
SELECT DISTINCT ?uri
```

```
WHERE {
```

```
    ?uri dbo:occupation dbr:Bandleader .
```

```
    ?uri dbo:instrument dbr:Trumpet .
```

```
}
```

Results

uri
http://dbpedia.org/resource/Brad_Turner_(musician)
http://dbpedia.org/resource/Jack_Jackson_(British_radio)
.....

Give me the number of all trumpet players that were bandleaders.

Select Queries

- Give me the number of all trumpet players that were bandleaders.

```
PREFIX dbo: <http://dbpedia.org/ontology/>
```

```
PREFIX dbr: <http://dbpedia.org/resource/>
```

```
SELECT COUNT (DISTINCT ?uri)
```

```
WHERE {
```

```
    ?uri dbo:occupation dbr:Bandleader .
```

```
    ?uri dbo:instrument dbr:Trumpet .
```

```
}
```

Results

46

Which countries have places with
more than two caves?

Select Queries

- Which countries have places with more than two caves?

```
PREFIX dbo: <http://dbpedia.org/ontology/>
```

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
```

```
SELECT DISTINCT ?uri
```

```
WHERE {
```

```
    ?cave rdf:type dbo:Cave .
```

```
    ?cave dbo:location ?uri .
```

```
    ?uri rdf:type dbo:Country .
```

```
}
```

```
GROUP BY ?uri
```

```
HAVING (COUNT(?cave) > 2)
```

uri
http://dbpedia.org/resource/Republic_of_Ireland
http://dbpedia.org/resource/Azerbaijan
http://dbpedia.org/resource/Jamaica
http://dbpedia.org/resource/Spain
...

Which software has been developed by organizations founded in California?

Select Queries

- Which software has been developed by organizations founded in California?

```
PREFIX dbo: <http://dbpedia.org/ontology/>
```

```
PREFIX dbp: <http://dbpedia.org/property/>
```

```
PREFIX dbr: <http://dbpedia.org/resource/>
```

```
SELECT DISTINCT ?uri
```

```
WHERE {
```

```
    ?company rdf:type dbo:Organisation .
```

```
    { ?uri dbo:developer ?company . }
```

```
    UNION
```

```
    { ?uri dbp:developer ?company . }
```

```
    ?uri rdf:type dbo:Software .
```

```
    ?company dbo:foundationPlace d
```

```
}
```

Result

uri
http://dbpedia.org/resource/AdWords
http://dbpedia.org/resource/Kylix_(software)
http://dbpedia.org/resource/Open64
http://dbpedia.org/resource/Picasa
...

Was Natalie Portman born in the
United States?

Ask Queries

- Was Natalie Portman born in the United States?

PREFIX dbo: <<http://dbpedia.org/ontology/>>

PREFIX dbr: <<http://dbpedia.org/resource/>>

ASK

WHERE {

dbr:Natalie_Portman dbo:birthPlace ?city .

?city dbo:country dbo:United_States .

}

Result

false

Is Rome the capital of Italy?

Ask Queries

- Is Rome the capital of Italy?

```
PREFIX dbo: <http://dbpedia.org/ontology/>
```

```
PREFIX dbr: <http://dbpedia.org/resource/>
```

```
SELECT COUNT (DISTINCT ?uri)
```

```
ASK
```

```
WHERE {
```

```
    dbr:Italy dbo:capital dbr:Rome .
```

```
}
```

Result

true

Does Elizabeth Tylor have a
birthdate in DBpedia?

Ask Queries

- Does Elizabeth Tylor have a birthdate in DBpedia?

```
PREFIX dbo: <http://dbpedia.org/ontology/>
```

```
PREFIX dbr: <http://dbpedia.org/resource/>
```

```
ASK
```

```
WHERE {
```

```
    dbr:Elizabeth_Taylor dbo:birthDate ?any .
```

```
}
```

Result

true

Is Milan a city in Germany?

Ask Queries

- Is Milan a city in Germany?

```
PREFIX dbo: <http://dbpedia.org/ontology/>
```

```
PREFIX dbr: <http://dbpedia.org/resource/>
```

```
ASK
```

```
WHERE {
```

```
    dbr:Milan dbo:country dbr:Germany .
```

```
}
```

Result

false

Is Michelle Obama the spouse of
Barack Obama?

Ask Queries

- Is Michelle Obama the spouse of Barack Obama?

PREFIX dbo: <http://dbpedia.org/ontology/>

PREFIX dbr: <http://dbpedia.org/resource/>

ASK

WHERE {

dbr:Barack_Obama dbo:spouse dbr:Michelle_Obama .

}

Result

true

Build a graph such that all entities heavier than 10000000000 are American basketball players

Construct Queries

- Build a graph such that all entities heavier than 1000000000 are American basketball players

```
PREFIX dbo: <http://dbpedia.org/ontology/>
```

```
PREFIX res: <http://dbpedia.org/resource/>
```

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
```

```
CONSTRUCT {
```

```
    ?x rdf:type dbo:AmericanFootballPlayer .
```

```
}
```

```
WHERE {
```

```
    ?x dbo:weight ?kg .
```

```
    FILTER (?kg > 1000000000)
```

```
}
```

Result

Construct Queries

- Build a graph such that 1000000000 are AmericanFootballPlayer

PREFIX dbo: <http://dbpedia.org/ontology/

PREFIX res: <http://dbpedia.org/resource/

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#

CONSTRUCT {

 ?x rdf:type dbo:AmericanFootballPlayer

WHERE {

 ?x dbo:weight ?kg .

 FILTER (?kg > 1000000000)

}

Result

Prefix	Namespace IRI
dbo	http://dbpedia.org/ontology/
n4	http://dbpedia.org/resource/Landkreuzer_P.
rdf	http://www.w3.org/1999/02/22-rdf-syntax-ns#
xsdh	http://www.w3.org/2001/XMLSchema#
dbr	http://dbpedia.org/resource/

Subject Item
dbr:Crawler-transporter
rdf:type
[dbo:AmericanFootballPlayer](http://dbpedia.org/resource/AmericanFootballPlayer)

Subject Item
n4:_1500_Monster
rdf:type
[dbo:AmericanFootballPlayer](http://dbpedia.org/resource/AmericanFootballPlayer)

Subject Item
dbr:Kinzua_Bridge
rdf:type
[dbo:AmericanFootballPlayer](http://dbpedia.org/resource/AmericanFootballPlayer)

Subject Item
dbr:Schwerer_Gustav
rdf:type
[dbo:AmericanFootballPlayer](http://dbpedia.org/resource/AmericanFootballPlayer)

Build a graph that associates the type **dbo:Aviator** to all those entities that have **dbr:airline** as occupation in DBpedia

Construct Queries

- Build a graph that attributes the type **dbo:Aviator** to all those entities that have **dbo:airline** as occupation in DBpedia

```
PREFIX dbo: <http://dbpedia.org/ontology/>
```

```
PREFIX dbr: <http://dbpedia.org/resource/>
```

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
```

```
CONSTRUCT {  
    ?x a dbo:Aviator.
```

```
}
```

```
WHERE {  
    ?x dbo:occupation dbo:Airline .
```

```
}
```

Result

Construct Queries

- Build a graph that attributes the type **dbo:Aviator** to all those entities that have **dbp:airline** as occupation in DBpedia

```
PREFIX dbo: <http://dbpedia.org/ontology/>
```

```
PREFIX dbp: <http://dbpedia.org/property/>
```

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
```

```
CONSTRUCT {  
    ?x a dbo:Aviator.
```

```
}
```

```
WHERE {
```

```
    ?x dbo:occupation dbp:airline
```

```
}
```

Result

Prefix	Namespace IRI
dbo	http://dbpedia.org/ontology/
dbp	http://dbpedia.org/property/
rdf	http://www.w3.org/1999/02/22-rdf-syntax-ns#
xsd	http://www.w3.org/2001/XMLSchema#
dbp	http://dbpedia.org/resource/

Statements

dbp:Juan_Trippe rdf:type [dbo:Aviator](http://dbpedia.org/ontology/Aviator) .

dbp:Walter_Varney rdf:type [dbo:Aviator](http://dbpedia.org/ontology/Aviator) .

dbp:Jeffrey_Skiles rdf:type [dbo:Aviator](http://dbpedia.org/ontology/Aviator) .

Describe all entities that have
dbr:airline as occupation

Describe Queries

- Please describe all people born in Milan

```
PREFIX dbo: <http://dbpedia.org/ontology/>
```

```
PREFIX dbr: <http://dbpedia.org/resource/>
```

```
DESCRIBE ?x
```

```
WHERE {
```

```
  ?x dbo:occupation dbo:Airline
```

```
}
```

RESULT

Describe Queries

■ Please describe all people born in Milan

Prefix Namespace IRI

dbpedia-de <http://de.dbpedia.org/resource/>

n36 <http://www.inc.com/30years/articles/>

xsdh <http://www.w3.org/2001/XMLSchema#>

n23 <http://data.bibliotheken.nl/id/thes/>

yago <http://dbpedia.org/class/yago/>

dbo <http://dbpedia.org/ontology/>

...

dbr:American_Overseas_Airlines dbo:wikiPageWikiLink dbr:Juan_Trippe .

dbr:Robert_Six dbo:wikiPageWikiLink dbr:Juan_Trippe , dbr:Walter_Varney .

dbr:List_of_burials_at_Green-Wood_Cemetery dbo:wikiPageWikiLink dbr:Juan_Trippe .

...

Please describe all subjects and objects of the **dbo:branchFrom** relation

Describe Queries

- Please describe all subjects and objects of the `dbo:branchFrom` relation

PREFIX `dbo:` `<http://dbpedia.org/ontology/>`

PREFIX `dbr:` `<http://dbpedia.org/resource/>`

DESCRIBE `?x ?y`

WHERE {

`?x dbo:branchFrom ?y`

}

RESULT

Exercise no. 7 // More SPARQL

- Exercises #5-#12 from “Knowledge Representation and Reasoning – Mod. 2 - Exercise Sheet 3 & 4: RDF & SPARQL” contain many more queries with specific features and solutions.
- **Solutions in the file**

References

- <https://www.w3.org/TR/rdf-sparql-query/>
- <https://www.w3.org/RDF/>
- <https://wiki.dbpedia.org/>

THANK YOU